Root-zero graceful labelings with a three-leaf star in a two-parameter diameter-six family

Abstract

Let $T(u, v)$ be the centrally rooted tree whose three branches have depth-two leaf multiplicities $(1, 1, 3)$, $(1, 1, 2v)$ and $(1, 2u)$, where u, v are positive integers. We give an explicit graceful labeling of $T(u, v)$ with the centre labeled zero. The construction first labels a merged tree with depth-one child counts $3, 3, 1$, reserves two prescribed pendant labels, moves one pair to create the three-leaf star, and then splits one star. Reflection pairs redistribute the remaining leaves while preserving all edge differences. Four symbolic cases and two explicit small witnesses cover every positive parameter pair. The base assignments are adapted from our earlier, narrower family; their validity at the shifted edge count is checked independently of its theorem statement.

1 The family and the construction

A *graceful labeling* of a tree with q edges is a bijection from its vertices to $[0, q]_{\mathbb{Z}}$ such that the absolute differences across its edges are $1, \ldots, q$, each occurring once. We call a labeling *root-zero* when it assigns zero to the specified root. All intervals in this paper contain integers. When labels are distinct, we use a vertex label also to denote its vertex.

For positive integers u, v, define $T(u, v)$ as follows. Its root r has three children. The first has three children with respectively $1, 1, 3$ pendant children. The second has three children with respectively $1, 1, 2v$ pendant children. The third has two children with respectively $1, 2u$ pendant children. There are no other vertices. Thus its branch leaf multiplicities are

$$(1, 1, 3), \qquad (1, 1, 2v), \qquad (1, 2u). \tag{1}$$

The order within each tuple and the order of the branches are immaterial. Write

$$S = u + v, \qquad q = 19 + 2S. \tag{2}$$

Indeed, there are three edges from the root, eight to depth two, and $8 + 2S$ to depth three. All leaves have depth three, and leaves in distinct branches have distance six. Consequently $T(u, v)$ has diameter six and unique centre r.

Theorem 1. *For every pair of positive integers u, v, the tree $T(u, v)$ has an explicit graceful labeling with r labeled zero.*

The proof uses the standard skeleton except on the line $u = 1$, $q \equiv 2 \pmod 3$, $q \geq 29$, where it uses an alternate skeleton. The two isolated exceptional parameter pairs have explicit assignments. Whenever a formula asks for disjoint reflection pairs, choose them in increasing order of their smaller labels.

The auxiliary *merged tree* $M(u, v)$ has branch leaf multiplicities

$$(1, 1, 3), \qquad (1, 1, 2v), \qquad (2u + 2). \tag{3}$$

It also has q edges: the change in the number of depth-two vertices is balanced by the change in the number of leaves. We label $M(u, v)$ with two particular leaves reserved at its exceptional depth-two vertex. One pair transfer creates the three-leaf star, further transfers adjust the other large star, and a local operation turns the last branch into $(1, 2u)$.

The earlier edition of this manuscript [1] treated $(1, 1, 1), (1, 1, 2v), (1, 2u)$, with $q = 17 + 2(u + v)$, and supplied the skeleton, anchor, and base leaf assignments adapted below. At the present edge count its old parameter sum would be $S_{\text{old}} = (q - 17)/2 = S + 1$. The old family theorem does not imply Theorem 1; the new transfer to the three-leaf star and its exact pair accounting are proved here. We reproduce the base rows and difference checks so that no unmentioned instance of the old theorem is required.

Relation to established results

Root-zero gracefulness of the merged family follows already from Superdock's Theorem 4(a) in [6, p. 5]. That theorem applies to complete diameter-six trees with odd child counts above the last two levels. For $M(u, v)$ these counts are 3 at the root and $3, 3, 1$ at depth one; every depth-two vertex has a child. The conclusion alone does not prescribe the two leaf labels and incidences required for our local operation. Those data are constructed below.

For $T(u, v)$, the depth-one child counts are instead $3, 3, 2$. Thus the common odd-child hypothesis of that theorem fails. The even and quasi-even degree-sequence classes of Balbuena, García-Vázquez, Marcote and Valenzuela [2] provide another sufficient-condition route. Our centre-neighbour degrees are mixed, $4, 4, 3$, and we do not use that route here. Neither comparison asserts that all other composition or transfer results are inapplicable.

Panda and Mishra's Theorem 2.3 [3, p. 273] allows one even-child vertex at depth one, but its additional conditions exclude (1). In its notation our tree has $m = 3$, $m_e = 1$, $m_o = 2$ and $n = n_e = n_o = 0$. Alternative (a) requires $m + n$ even; alternative (b) requires either $n_e \equiv 0 \pmod 4$ with $n_o \geq 1$, or $n_e \equiv 1 \pmod 4$. Neither holds. Its common uniform-parity restriction on lower stars also fails in the two branches with an enlarged star.

The component-moving class of Mishra and Panda [4] requires each diameter-four branch centre to have an odd number of depth-two children. The exceptional branch of $T(u, v)$ has two. The degree hypotheses of the generalized classes in [5, Theorems 3.1–3.3] likewise do not directly apply to the centre-neighbour degrees $4, 4, 3$. These are bounded hypothesis comparisons, not a classification of all known diameter-six graceful families.

The cited sufficient conditions do not directly yield our specified root-zero family. We make no literature-wide priority claim; our purpose is the explicit all-parameter construction and its verifiable anchor and transfer accounting.

2 Two operations and the standard skeleton

Lemma 2 (Pair transfer)**.** *Suppose x, y are distinct pendant vertices at a vertex b of a gracefully labeled tree, and p is another vertex distinct from x, y. If $x + y = b + p$, deleting bx, by and adding px, py preserves gracefulness.*

Proof. The vertex labels do not change, and

$$|p - x| = |b - y|, \qquad |p - y| = |b - x|.$$

Thus the two affected differences exchange places. Removing the pendant vertices leaves a tree; reattaching each at p again gives a tree. □

This is the two-leaf instance of the established transfer sum condition; see [6, Section 1.1, p. 3]. We only use the elementary form above, on disjoint pairs of pendant vertices.

Lemma 3 (Splitting a star)**.** *Suppose a gracefully labeled tree contains an edge AB and pendant edges BC, BZ, where the four labels are distinct and, for a nonzero integer d,*

$$A = C - 2d, \qquad B = C + d, \qquad Z = C - d. \tag{4}$$

Deleting BC, BZ and adding AC, CZ preserves gracefulness. If A is the parent of B, the two pendant vertices are replaced by a new child C of A with sole child Z.

Proof. The deleted edges have differences $|d|, 2|d|$; the added edges have differences $2|d|, |d|$. Removing the leaves C, Z leaves a tree containing AB. Adding the path A, C, Z therefore gives a tree on the same vertex set. Its other edges and every vertex label are unchanged. □

In $M(u, v)$ let B denote the depth-two vertex of the singleton branch and A its parent. We will keep leaves C, Z at B until applying Lemma 3. We call (A, B, C, Z) the *anchor*. Before the split, B has $2u + 2$ leaves and one of the other six depth-two vertices, denoted P, has $2v$ leaves. Among the other five depth-two vertices, Q, the recipient of the first transfer, has three leaves and the remaining four each have one. The split then gives exactly (1).

The standard skeleton has root label 0, depth-one labels $q, 1, q - 1$, and the depth-two groups in Table 1. Only the placement of the singleton group varies with the residue of q.

q mod 3	Children of q	Children of 1	Children of $q - 1$
0	$2, q - 2, 3$	$q - 3, 4, q - 4$	5
1	$2, q - 2, 3$	$q - 3$	$4, q - 4, 5$
2	2	$q - 2, 3, q - 3$	$4, q - 4, 5$

Table 1: The standard skeleton. Each displayed group is attached to its column's depth-one vertex.

The eleven labels used by this skeleton are $[0, 5]_{\mathbb{Z}} \cup [q - 4, q]_{\mathbb{Z}}$. The three root-edge differences are $q, 1, q - 1$. In each row of Table 1, the seven remaining differences form the multiset

$$\{\{2, 3, q - 2, q - 3, q - 4, q - 5, q - 6\}\}.$$

For example, in the first row they occur as $q - 2, 2, q - 3, q - 4, 3, q - 5, q - 6$, in the displayed group order. It follows that the unused vertex labels and required leaf differences are

$$I = [6, q - 5]_{\mathbb{Z}}, \qquad D = [4, q - 7]_{\mathbb{Z}}. \tag{5}$$

Both have cardinality $q - 10 = 2S + 9$. Thus it remains to assign each label in I to one depth-two parent, using every difference in D exactly once and meeting the leaf counts stated above.

Each row of Table 2 satisfies (4) by substitution, and $d \neq 0$ for $q \geq 23$. The labels C, Z are integers in their stated residue classes. Their interval membership and distinction from each other and from occupied leaf labels will be checked in each construction. Since A, B lie outside I, these checks also ensure that the four anchor vertices are distinct.

$q \bmod 3$	A	B	C	Z	$d = B - C$
0	$q-1$	5	$(q+9)/3$	$(2q+3)/3$	$(6-q)/3$
1	1	$q-3$	$(2q-5)/3$	$(q-1)/3$	$(q-4)/3$
2	q	2	$(q+4)/3$	$(2q+2)/3$	$(2-q)/3$

Table 2: The prescribed standard anchor.

Parent p	Leaf x	$\lvert x-5\rvert$	$\lvert x-p\rvert$
2	$q-5$	$q-10$	$q-7$
$q-2$	6	1	$q-8$
3	$q-6$	$q-11$	$q-9$
$q-3$	7	2	$q-10$
4	$q-7$	$q-12$	$q-11$
$q-4$	8	3	$q-12$
$q-4$	h	$h-5$	$h-5$

Table 3: The seven reassigned leaves for $q \equiv 0 \pmod 3$.

3 Standard construction for $q \equiv 0 \pmod 3$

Assume first $q = 27 + 6t$ for an integer $t \geq 0$, and put $h = (q+1)/2$. Use the first skeleton and anchor rows. Assign the seven leaves in Table 3 to the indicated ordinary parents, and assign every other label of I to $B = 5$.

For completeness, all labels needed for the distinctness and anchor checks occur in the following strictly increasing list:

$$6 < 7 < 8 < C-3 < C < h < Z-3 < Z < q-7 < q-6 < q-5. \tag{6}$$

Indeed, the middle terms $C-3, C, h, Z-3, Z$ are respectively $9+2t, 12+2t, 14+3t, 16+4t, 19+4t$, and $q-7 = 20+6t$. Every consecutive difference is positive. This proves that the seven assigned labels are distinct and lie in I, and that C, Z remain at B.

If all labels of I were assigned to 5, their differences would be $[1, q-10]_{\mathbb{Z}}$. The last two columns of Table 3 show that the reassignment replaces $1, 2, 3$ by $q-9, q-8, q-7$, with all other differences canceled as multisets. The resulting leaf differences are exactly D. Thus the displayed assignment and the skeleton form a gracefully labeled tree. At this stage $P = q-4$ has two leaves, B has $|I| - 7 = 2S+2$ leaves, and every other depth-two parent has one. Set $Q = q-2$, a parent in the other ordinary root branch.

Consider the reflection $\rho(x) = q+1-x$ of I, whose fixed point is h. The seven labels removed from B are the three complete reflection pairs

$$(6, q-5), \quad (7, q-6), \quad (8, q-7),$$

and the fixed point h. The remaining $2S+2$ labels at B therefore form $S+1$ disjoint two-element pairs. The mates of C, Z are respectively $Z-3, C-3$, as follows from $C + Z = q+4$. By (6), these two anchor pairs are intact, distinct, and contain no fixed point.

For $t \geq 1$, the pair $(C-3, Z+2)$ lies at B by (6): $C-3$ lies strictly between 8 and C, and $Z+2 = 21+4t < q-7 = 20+6t$. Its sum is $q+3 = B+Q$. Transfer it to Q, which now has three

leaves. This move breaks the reflection pair $(C-3, Z)$ containing an anchor and the non-anchor pair $(C-5, Z+2)$; the other anchor pair $(C, Z-3)$ remains intact and excluded. Thus $S-2$ complete, non-anchor pairs remain at B. The inequalities above ensure these are distinct available labels for every $t \geq 1$.

The sum of the donor and recipient labels for the remaining transfers is $B+P = 5+(q-4) = q+1$. Hence Lemma 2 allows any of those pairs to be transferred to P. Transfer $v-1$ of them; this is possible because $v-1 \leq S-2$ for $u \geq 1$. Afterwards

$$\deg_{\text{leaf}}(P) = 2+2(v-1) = 2v, \qquad \deg_{\text{leaf}}(B) = 2S-2(v-1) = 2u+2.$$

The reserved leaves C, Z are still at B. Lemma 3 now gives the required root-zero labeling of $T(u, v)$ for $t \geq 1$.

At $t = 0$ $(q = 27)$, $Z+2 = q-6$ was already reassigned and the preceding move is unavailable. Instead transfer $(13, 17)$ from $B = 5$ to $Q = 25$; their sum is $30 = B+Q$. One complete non-anchor reflection pair remains at B, sufficient for $(u, v) = (2, 2)$ and $(3, 1)$, whose required numbers of further transfers are one and zero. The remaining pair $(1, 3)$ is handled by the exact witness in Table 5.

4 Standard construction for $q \equiv 1 \pmod 3$

Here $q = 25+6t$, $t \geq 0$. Use the second standard skeleton and anchor rows. We start with four leaves at $B = q-3$ and assign all but nine labels of I to $P = 4$. The exceptions are Table 4.

Parent p	Leaf x	$\lvert x-4\rvert$	$\lvert x-p\rvert$
B	C	$C-4$	$Z-1$
B	Z	$Z-4$	$C-1$
B	7	3	$q-10$
B	$C+3$	$C-1$	$Z-4$
2	$q-5$	$q-9$	$q-7$
$q-2$	6	2	$q-8$
3	$q-6$	$q-10$	$q-9$
$q-4$	$Z+2$	$Z-2$	$C-4$
5	$Z+3$	$Z-1$	$Z-2$

Table 4: The nine exceptions to assignment at $P = 4$ for $q \equiv 1 \pmod 3$.

Put $h = (q+1)/2$. The following list is strictly increasing:

$$6 < 7 < Z < Z+2 < Z+3 < h < C < C+1 < C+3 < q-6 < q-5. \tag{7}$$

To verify it, substitute $Z = 8+2t$, $h = 13+3t$, $C = 15+4t$, and $q-6 = 19+6t$. The successive gaps are positive for $t \geq 0$. In particular every exception is a distinct label of I, while $h, C+1$ remain at P.

The provisional assignment of all I to $P = 4$ gives the interval of differences $[2, q-9]_{\mathbb{Z}}$. Comparing the last two columns of Table 4, the only uncanceled change removes $2, 3$ and adds $q-8, q-7$. Hence the final differences are exactly D. The resulting tree is graceful, with four leaves at B, $q-19 = 2S$ leaves at P, and one leaf at every other depth-two parent. Both anchors are already at B. Let $Q = q-2$, in the other ordinary root branch.

For the reflection $\rho(x) = q + 1 - x$ of I, the nine exceptions consist of four full pairs

$$(6, q-5), \quad (7, q-6), \quad (Z, C+3), \quad (C, Z+3) \tag{8}$$

and the single label $Z+2$. Its mate is $C+1$, which remains at P, as does the fixed point h. The full pairs in (8) follow from $C + Z = q - 2$; the same identity gives $(Z+2) + (C+1) = q+1$. There are initially $(|I|-1)/2 = S+4$ two-element pairs. Removing the four full pairs and hitting one more leaves $S-1$ full pairs at P, together with the two unpaired labels $C+1, h$. Distinctness of all these exceptional labels follows from (7).

The labels $h, h+1$ remain at P. Their sum is $q+2 = P+Q$, so Lemma 2 moves them to Q, giving it three leaves. The fixed point h is consumed; $h+1$ is one member of the full pair $(h-1, h+1)$. Exactly $S-2$ full pairs remain at P.

Since $P + B = 4 + (q-3) = q+1$, transfer $u-1$ of those pairs from P to B. The inequality $u - 1 \le S - 2$ is exactly $v \ge 1$. The new leaf counts are

$$\deg_{\text{leaf}}(B) = 4 + 2(u-1) = 2u+2, \qquad \deg_{\text{leaf}}(P) = 2S - 2 - 2(u-1) = 2v.$$

The leaves C, Z were at B throughout. Apply Lemma 3 to obtain the required labeling.

5 Standard construction for $q \equiv 2 \pmod 3$, $u \ge 2$

Let $q = 29 + 6t$, $t \ge 0$, and $m = (q-1)/2$. Use the third standard skeleton and anchor rows. Give the following leaves their indicated parents, and every other label of I to $B = 2$:

p	3	$q-3$	$q-2$	4	5	$q-4$
leaves	$q-6$	$q-7,\ m-2$	6	m	$m+2$	$m-1.$

(9)

Thus the ordinary parent with two leaves is $P = q - 3$.

All relevant distinctness checks follow from the strictly increasing list

$$\begin{aligned} 6 < C-3 < C < m-2 < m-1 < m < m+1 < m+2 \\ < Z-3 < Z < q-7 < q-6 < q-5. \end{aligned} \tag{10}$$

Here $C = 11 + 2t$, $m = 14 + 3t$, $Z = 20 + 4t$, and $q - 7 = 22 + 6t$. In particular the gap between C and $m-2$ is $1+t$, as is the gap between $m+2$ and $Z-3$; all other gaps are also positive. The seven reassigned labels therefore lie in I, are distinct, and avoid both anchor labels and their prospective mates.

Assigning all of I to $B = 2$ would already give the required difference interval D. Associate an old difference d with the unique label $d+2$. Reassigning a leaf x to p therefore replaces its old label index by

$$F_p(x) = 2 + |x - p|.$$

The seven reassignments in (9) form two disjoint permutation cycles on the label indices:

$$q-6 \xrightarrow{p=3} q-7 \xrightarrow{p=q-3} 6 \xrightarrow{p=q-2} q-6, \tag{11}$$

$$m \xrightarrow{p=4} m-2 \xrightarrow{p=q-3} m+2 \xrightarrow{p=5} m-1 \xrightarrow{p=q-4} m. \tag{12}$$

These identities follow by substituting $q = 2m+1$; all absolute-value signs are fixed by (10) and by the parent labels lying outside I. Consequently the reassignments permute the old differences and

preserve D. We have a graceful tree with two leaves at P and $2S+2$ leaves at B. Set $Q = q-4$ in the other ordinary root branch.

Now use $\rho(x) = q-1-x$, with fixed point m, on the subinterval $J = [6, q-7]_{\mathbb{Z}}$. This interval initially has $S+3$ two-element pairs and the fixed point. Its two complementary labels in I are $q-6, q-5$. The seven labels removed from B consist of $q-6$, m, the two full pairs $(6, q-7)$ and $(m-2, m+2)$, and the singleton $m-1$. Thus three full pairs of J are hit, leaving S full pairs at B. Its other two leaves are the unpaired labels $m+1$ and $q-5$.

The reflection mates of C, Z are $Z-3, C-3$, respectively, because $C+Z = q+2$. By (10), both full anchor pairs remain at B and are distinct.

The labels $(C-3, Z-1)$ lie at B: the first lies between 6 and C, and $Z-1$ lies strictly between $m+2$ and Z, so neither belongs to (9). Their sum is $q-2 = B+Q$; transfer this pair to Q to create its three leaves. This breaks the anchor pair $(C-3, Z)$ and the non-anchor pair $(C-2, Z-1)$. Excluding the other anchor pair $(C, Z-3)$ leaves exactly $S-3$ complete non-anchor pairs at B. Since $B+P = 2+(q-3) = q-1$, these are transfer pairs for B, P. For $u \geq 2$,

$$v-1 \leq S-3.$$

Transfer $v-1$ such pairs from B to P. Their counts become $2u+2$ and $2v$, respectively. Splitting the unchanged anchor completes this case.

6 An alternate skeleton on the remaining line

Suppose $u = 1$, $q = 29+6t$ with $t \geq 0$. Then $q = 21+2v$, so these parameters are exactly $v = 4+3t$. Give the root label 0 and use the following depth-one labels and depth-two groups:

depth-one label	q	2	$q-1$
depth-two children	$3, 4, q-4$	$q-3$	$1, 5, q-2.$

(13)

The eleven skeleton labels are still $[0,5]_{\mathbb{Z}} \cup [q-4, q]_{\mathbb{Z}}$. The root-edge differences are $q, 2, q-1$; the other seven are $q-3, q-4, 4, q-5, q-2, q-6, 1$, in displayed group order. Thus the skeleton differences are $\{1,2,4\} \cup [q-6, q]_{\mathbb{Z}}$, and the required leaf differences are

$$D' = \{3\} \cup [5, q-7]_{\mathbb{Z}}. \tag{14}$$

Use the anchor

$$A = 2, \qquad B = q-3, \qquad C = \frac{2q-4}{3}, \qquad Z = \frac{q+1}{3}, \qquad d = \frac{q-5}{3}. \tag{15}$$

These satisfy (4). Give nine labels their parents as follows, assigning every other label of I to $P = 4$:

p	B	1	$q-2$	5	3	$q-4$
leaves	$C, Z, 7, Z+2$	$q-6$	6	8	$Z-1$	$C+2.$

(16)

The increasing list

$$6 < 7 < 8 < Z-1 < Z < Z+2 < C < C+2 < q-6 < q-5 \tag{17}$$

follows from $Z = 10+2t$, $C = 18+4t$, and $q-6 = 23+6t$. It proves all interval and distinctness requirements for the nine assigned labels, including the reserved leaves C, Z.

The all-at-P provisional differences are $[2, q-9]_{\mathbb{Z}}$. Four rows of (16) have the following effect:

p	x	$\lvert x-4\rvert$	$\lvert x-p\rvert$
1	$q-6$	$q-10$	$q-7$
$q-2$	6	2	$q-8$
B	7	3	$q-10$
5	8	4	3.

(18)

Their net change removes $2, 4$ and adds $q-8, q-7$. For the remaining five rows identify a difference with its old label index by adding 4. Under $G_p(x) = 4 + |x-p|$, these rows are precisely the cycles

$$C \xrightarrow{p=B} Z+2 \xrightarrow{p=B} C, \tag{19}$$

$$Z \xrightarrow{p=B} C+2 \xrightarrow{p=q-4} Z-1 \xrightarrow{p=3} Z. \tag{20}$$

For instance $q+1-C = Z+2$ and $q-Z = C+2$; the other equalities follow from the same formulas. All signs are valid by (17). The cycles therefore preserve the corresponding old differences, and the complete leaf difference set is exactly D'.

The resulting merged tree is graceful. Its parent B has four leaves, including C, Z; P has $|I|-9 = q-19 = 2v+2$ leaves; the five other depth-two parents have one each. Set $Q = q-2$ in the other ordinary root branch. Put $m = (q-1)/2 = 14+3t$. The two labels $m, m+3$ are strictly between $Z+2 = 12+2t$ and $C = 18+4t$, so they are outside the nine exceptions in (16) and remain at P. Their sum is $q+2 = P+Q$. Transfer them to Q, leaving $2v$ leaves at P and three at Q. Apply Lemma 3 using (15); the exceptional branch becomes $(1, 2)$. This proves the required labeling on the entire $u = 1$ line of this residue class. The first member is $(u, v) = (1, 4)$ at $q = 29$.

7 Small witnesses and completion of the proof

For $q = 23$ the only positive pair is $(u, v) = (1, 1)$. At $q = 27$ the case $(1, 3)$ is the sole pair not covered by the standard $q \equiv 0 \pmod 3$ transfer above. Both have explicit merged-tree assignments in Table 5. A cell $p : x_1, \ldots, x_k$ means that each displayed leaf label is attached to parent p. All unlisted edges are those of the applicable standard skeleton in Table 1.

$(u, v); q$	Parent-to-leaf assignments and anchor
$(1, 1); 23$	$2 : 9, 13, 16, 18$; $3 : 7$; $4 : 10, 12, 17$; $5 : 15$; $19 : 14$; $20 : 8, 11$; $21 : 6$. $(A, B, C, Z) = (23, 2, 9, 16)$; $P = 20, Q = 4$.
$(1, 3); 27$	$2 : 22$; $3 : 18$; $4 : 21$; $5 : 9, 12, 16, 19$; $23 : 10, 11, 13, 14, 15, 17$; $24 : 8$; $25 : 6, 7, 20$. $(A, B, C, Z) = (26, 5, 12, 19)$; $P = 23, Q = 25$.

Table 5: Exact small merged-tree witnesses. In each row the anchor leaves C, Z are subsequently split by Lemma 3.

For $(1, 1)$, the displayed labels partition $I = [6, 18]_{\mathbb{Z}}$. Their absolute differences at the listed parents are exactly $D = [4, 16]_{\mathbb{Z}}$; the parent counts are $4, 2, 3$ at B, P, Q and one at each other depth-two parent. The anchor identities hold with $d = -7$. For $(1, 3)$, the displayed labels partition $I = [6, 22]_{\mathbb{Z}}$, the differences are exactly $D = [4, 20]_{\mathbb{Z}}$, and the counts at B, P, Q are $4, 6, 3$. Its anchor has $d = -7$. These direct finite checks, together with the skeleton check, prove that both merged trees are graceful before the split. The portable checker in §8 records the rows and independently tests all resulting edges, labels, differences, and rooted branch profiles.

Proof of Theorem 1. The edge count $q = 19 + 2S$ is odd and at least 23. If $q \equiv 0 \pmod 3$, then $q = 27 + 6t$. Section 3 handles $t \geq 1$ and, at $t = 0$, the pairs $(2, 2), (3, 1)$; Table 5 handles $(1, 3)$. If $q \equiv 1 \pmod 3$, then $q = 25 + 6t$, and Section 4 applies. If $q \equiv 2 \pmod 3$, Table 5 handles $q = 23$; for $q \geq 29$, Section 5 applies when $u \geq 2$ and Section 6 when $u = 1$. These cases exhaust all positive pairs.

Each base uses every label and leaf difference exactly once. The first pair transfer gives exactly three leaves at Q; any remaining transfers give exactly $2v$ leaves at P and $2u + 2$ at B while preserving the anchor leaves. Lemma 2 preserves all differences, and Lemma 3 turns the exceptional merged branch into $(1, 2u)$. The root remains labeled zero, and the resulting branch multiplicities are precisely (1). □

8 Formal and computational companions

The adjacent `lean/` project formalizes Theorem 1 for every positive parameter pair, including the two exceptional assignments. It supplies a computable labeling and proves its correspondence with the manuscript's original and merged trees, the local split, and the printed anchor and transfer formulas. It also proves the diameter-six and unique-centre statements. Its `README.md` gives the pinned build instructions and a guide to the declarations. The finite computations below provide separate supplementary checks of the displayed construction.

The computational companion distributed alongside this manuscript in `computational/` supplies the following program, finite input data and run instructions. The standard-library Python script `support/verify_extension.py` reconstructs every base assignment and subsequent transfer from the displayed formulas. It embeds neither a search answer nor an invocation of the earlier family theorem. The adjacent `support/small_solutions.json` gives the exact leaf rows of Table 5. For each pair it checks the final vertex labels $0, \ldots, q$, all edge differences $1, \ldots, q$ exactly once, q edges, the zero root, and the three branch profiles. Its default bounded run checks $1 \leq u, v \leq 100$ (10,000 pairs), including every symbolic case and the two exceptions. This corroborates formula-to-code correspondence; the universal argument is the preceding proof. The additional `support/check_construction.py` checks 70 displayed base leaf-to-parent entries at $t = 0$ and 14 boundary constructions, recording the base, first-transfer, redistributed and split graphs. Its finite inputs are in `support/formula_boundaries.json`. Repeated parent values denote different leaves of a star; the final vertex labels and edge differences are checked separately. For each literal small witness the first three stages coincide. These additional checks do not replace the all-parameter construction proof.

References

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